DATA ANALYTICS / LINKEDIN PROJECT CASE STUDY

ONLINE RETAIL SALES INTELLIGENCE

MySQL case study: revenue, customer behavior and product performance



$8.40M	22,190	4,372
Reported revenue	Orders	Customers

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Executive Overview

Project objective, analytical scope and methods

Objective

Transform transactional retail data into useful business intelligence using SQL queries, KPI analysis and structured reporting.

Scope

Revenue performance, customer purchase patterns, product demand, geographical sales contribution and operational recommendations.

Methods

MySQL queries, data cleaning, calculated revenue, grouping, aggregation, customer/product segmentation and business interpretation.

Business application

Use the results to prioritize customer retention, product inventory planning and market-specific decisions.

Retail Performance Snapshot

Reported statistics from the analyzed dataset

$8.40M

Revenue

22,190

Orders

4,372

Unique customers

406,431

Transaction records

Metric detail

The source reports total revenue of $8,401,284.13. These are analytical dataset figures, not client earnings or verified commercial outcomes.

Data Model and Preparation

Relational structure and validation workflow

Relational structure

The report identifies customers, products, orders and order_details as its core logical entities. Order details connect transactions with products.

Data preparation

Review missing values, check date consistency and duplicates, and calculate line revenue as Quantity × UnitPrice.

Analytical workflow

Understand the data; prepare and validate fields; aggregate revenue by month, country, product and customer; interpret the outputs.

SQL techniques

Filtering, grouping, aggregation, distinct counts and sorting support revenue and customer intelligence.

SQL Analysis Methods

Illustrative country-level revenue aggregation

```
SELECT Country,
    SUM(Revenue) AS Total_Revenue,
    COUNT(DISTINCT CustomerID) AS Customers
FROM final_retail_data
WHERE Country IS NOT NULL
GROUP BY Country
ORDER BY Total_Revenue DESC;
```

This query illustrates the aggregation approach described in the source report; it is not claimed to be a verbatim executable extract of the original project.

Additional tasks: distinct order counts, total revenue, top customers, purchase frequency, best-selling products and monthly sales trends.

RETAIL / CASE STUDY

Sales and Customer Insights

Market contribution and retention opportunities

6,861,884.20

UK revenue reported

$8.40M

Total revenue

Geographic concentration

The United Kingdom is identified as the largest revenue-generating country in the analyzed dataset.

Seasonality

Monthly revenue aggregation shows changes across the reporting period. The source report does not provide month-by-month figures in its narrative summary.

Customer intelligence

Customer-level SQL aggregation ranks buyers by revenue and purchase frequency to identify high-value and repeat purchasers.

Business Recommendations

Practical applications and analytical limitations

- Segment high-value and frequent customers for retention campaigns.
- Review the revenue concentration of leading countries and products.
- Use monthly revenue patterns to support inventory and marketing planning.
- Validate revenue definitions and underlying transaction records before operational decisions.
- Monitor purchase frequency and repeat-order behavior over time.

Portfolio note: insights and figures are based on the supplied project report. No new database execution, independent raw-data verification or causal modeling is claimed.